

DevGems — Quick Guide

DevGems is a web app to back up databases (PostgreSQL, MySQL/MariaDB, SQL Server, Oracle, MongoDB, SQLite), keep the backups **encrypted**, **verify** them, and **restore** them — on demand or on a schedule.

The 3 things it does

Step	What happens
1. Back up	Runs the DB's native dump tool and streams the output through AES-256-GCM straight into an encrypted <code>.enc</code> file. Plaintext never hits the disk.
2. Verify	Every backup is auto-checked: decryption proves it's intact + key is correct , and header/footer markers prove the dump is complete (not truncated). Shown as ✓ Verified.
3. Restore	Decrypts and loads a backup into a target database you choose (never the live source).

Key ideas in 60 seconds

- **One secret protects everything:** `BACKUP_ENCRYPTION_KEY` (in `.env.local`) is turned into an AES-256 key that encrypts **all backup files** and **all stored DB passwords**. Steal the files → useless without the key. **Lose the key → backups are gone forever.** Keep a copy in a password manager.
- **Login is passwordless:** you sign in with a key file (ECDSA). The private key never leaves your device.
- **Permissions:** admins do everything; members get fine-grained rights (view / run / edit / delete / manage users).
- **Everything streams:** dumps, verification, and restores never load the whole DB into memory — so multi-GB databases work fine.

Everyday use

Add a source: Connections → New → enter engine, host, port, database, user, password → Save. Hit **Test** to confirm it connects.

Back up: open the connection → **Backup now**, or set a **cron schedule** for automatic backups.

Keep only the latest few: set **Retention = 2 or 3** on the connection; older backups are pruned automatically.

Check a backup: the Backups page shows the file name, date/time, size, status, and a ✓ Verified / ✗ Unverified badge.

Restore: Backups → **Restore** → enter a *target* DB (e.g. `localhost` / a scratch database) → tick "recreate" + "auto-create roles" → **Restore now**.

Common gotchas

Problem	Fix
<code>pg_dump: server version mismatch</code>	Your dump tool is older than the DB. Install a matching client on the app host (<code>bash install-pg15-client.sh</code>). Nothing on the DB server changes.
Restore: <code>role "..."</code> does not exist	Keep Auto-create roles ticked (creates it for you).
Restore: <code>database already exists</code>	Tick Recreate target database (drops & recreates for a clean restore).
Backup failed , 0 bytes	Wrong DB name / credentials — the app refuses to save an empty dump. Fix and retry.

Command-line helpers (optional)

```
./dbbackup.sh                # guided, version-aware backup
node scripts/verify-backup.cjs --all      # verify every backup
node scripts/restore-backup.cjs --out d.sql # just decrypt a backup to a .sql file
node scripts/prune-backups.cjs --keep 3 --set --apply # keep newest 3
```

*Full details: see **DOCUMENTATION.md**.*